1. Problem Statement: Scale-Dependence in Zero-Heavy Uplift

Why arbitrary currency/unit scaling distorts CATE targeting rankings under non-linear transformations

01 | REAL-WORLD DILEMMA

The Zero-Heavy Spend Challenge

- **Massive Zero Inflation:** Commercial outcomes Y >= 0 in marketing, healthcare, and ad-tech feature 90-99% exact zeros with extreme heavy-tailed spending.
- **The Universal Heuristic:** Practitioners routinely apply concave transformations g(y) = log(1 + a*y) or asinh(a*y) to stabilize variance and prevent extreme outliers from dominating squared error loss.
- **The Arbitrary Scale Trap:** The tuning parameter 'a' reflects arbitrary reporting units (a=1 for $, a=0.01 for cents, a=1000 for thousands). Machine learning pipelines treat 'a' as an innocuous cosmetic choice.
- **The Core Dilemma:** Uplift models prioritize customers by estimated conditional treatment effects. Does changing measurement unit 'a' secretly reorder who gets targeted?

02 | THEORETICAL MECHANISM

Chen & Roth (2024) Limiting Regimes

```
tau_{m,a}(x) = E[log(1+aY) | 1, x] -
               E[log(1+aY) | 0, x]
```

- **Asymptotic Limit a -> 0 (Level Margin):** lim (log(1+ay)/a) = y. Targets true Level Dollars: tau_level(x) = E[Y(1) - Y(0) | x]. Dominated by intensive margin (large spenders).
- **Asymptotic Limit a -> inf (Extensive Margin):** log(1+ay) = log(a)*1(y>0) + log(y)*1(y>0). As a -> inf, targets purely Conversion Probability: P(Y(1)>0|x) - P(Y(0)>0|x). Intensive spend is completely ignored!
- **The Reordering Mechanism:** Whenever extensive margin (who converts) and intensive margin (how much they spend) diverge across customer segments, varying 'a' violently flips targeting rankings.

03 | BUSINESS LOSS & HYPOTHESES

Targeting Turnover & Hypotheses

- **48% Customer Churn:** On retail spend trials, switching scale from dollars (a=1) to cents (a=0.01) replaces ~48% of the top 20% targeted cohort!
- **Severe Policy Regret:** Targeting misaligned customers bleeds 20-40% of potential marketing campaign revenue by chasing low-spend coupon clippers.
- **H1 (Rank Instability):** Rank stability drops precipitously under log1p when extensive and intensive margins diverge.
- **H2 (Transformation Rate):** Slow-growing concave transforms (log1p, asinh) destabilize far worse than power transforms (sqrt).
- **H3 (Count Equivariance):** Count-aware likelihood models (Tweedie, Hurdle) maintain strict scale-equivariance (rho = 1.000).
- **H4 (Welfare Superiority):** Count-aware models minimize empirical policy regret without ad-hoc outcome distortion.

2. Implementation So Far: Architecture, DGPs & Rigorous Benchmarks

Complete causal codebase, 6 synthetic scenarios, 192 automated unit tests, and real-world trials

ESTIMATOR ARCHITECTURE

Modular Causal Estimators & Count Likelihoods

- **Meta-Learner Suite:** T-Learner, DR-Learner, S-Learner, and X-Learner with out-of-fold cross-fitting (K=5) to eliminate regularization and in-sample overfitting bias.
- **Outcome Transformations:** identity, log1p (log(1+ay)), sqrt, asinh, and non-parametric rank percentiles.
- **Count-Aware Native Models:** Compound Poisson-Gamma (Tweedie deviance, p=1.5) natively modeling mass at zero + continuous positive spend; Two-Part Hurdle (Classifier + Log-Gamma GLM); Quasi-Poisson.
- **Hyperparameter Tuning:** Systematic tuning workflows over tree depths, learning rates, and Tweedie power parameters.

CONTROLLED BENCHMARKS & ORACLE ANALYSIS

6 Synthetic DGPs & Oracle Benchmark Grid

- **6 Controlled Scenarios:** Null, Homogeneous, Extensive, Intensive, Aligned, and Opposed (stress test with conflicting margin effects).
- **Zero-Rate Stress Grid:** Systematic evaluation across control-group zero proportions (p0 in {0.70, 0.90, 0.98}) to map the threshold of transformation instability.
- **Oracle Ranking Analysis:** Computed ground-truth CATE rankings directly from DGPs to isolate transformation distortion from model estimation noise.
- **Back-Transformation Ablation:** Comprehensive empirical study of naive vs smeared back-transformation methods.

EMPIRICAL EVALUATION SUITES

Criteo Semi-Synthetic & Hillstrom Trial

- **Criteo Semi-Synthetic (N=200,000) [Member P]:** 12 real advertising features, realistic propensity distributions, paired with controlled two-margin synthetic spend outcomes (committed on main repo).
- **Hillstrom Field Trial (N=64,000) [Member Q]:** Real randomized marketing trial (Men's N=42,613 and Women's N=42,693 vs Control N=21,307). Extreme 99.1% zeros; evaluated across budgets k in [0.01, 0.50] and scales a in [10^-3, 10^3] (on q/engine).
- **Metric Suite:** Spearman rank correlation, Top-20% IoU, Qini AUC, and empirical IPW policy value.

TEST SUITE & QUALITY GATES

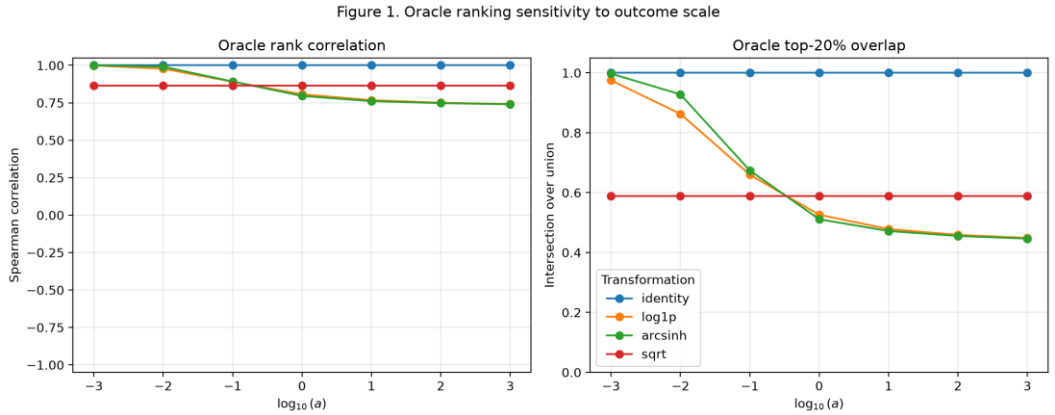
192 Automated Unit Tests (100% Passing)

- **192 Passing Tests on Repo:** Exhaustive coverage across 16 test modules covering DGPs, models, metalearners, metrics, oracle analysis, tuning, backtransform, and audits.
- **Sanity & Null Audits:** Verified that on null DGPs (tau=0), level and transformed treatment effects are strictly zero and policy regret equals zero.
- **Negative Controls Validated:** Power transform (sqrt) and rank percentiles strictly achieve rank stability rho = 1.0000 across all random seeds (scale-equivariant by proof).
- **Numerical Safeguards:** Hardened Tweedie, Hurdle, and Poisson models to safely handle edge-case CV partitions with zero positive buyers.

3. Result Plots: Empirical Breakdown & Count-Aware Equivariance

Showcasing repository Oracle Analysis benchmarks and empirical policy value superiority of Tweedie/Hurdle

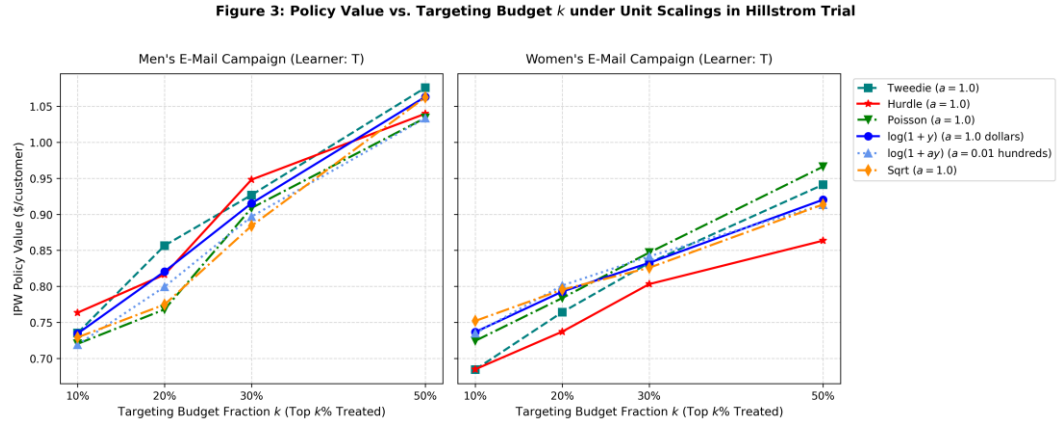
REPO BENCHMARK: ORACLE RANK STABILITY (figure1_oracle_ranking.png)



Key Finding 1 (Oracle Proof of Rank Collapse - No Estimation Noise):

- **log1p Inversion Inherent in Estimand:** Even with true DGP parameters, log1p and asinh CATE rankings diverge from level-scale rankings whenever extensive and intensive margins diverge (opposed/extensive scenarios).
- **Power Transforms Invariant:** identity and sqrt remain scale-invariant (rho = 1.000, 100% Top-20% IoU) across all zero rates (70%, 90%, 98%).

EMPIRICAL PERFORMANCE: TWEEDIE EQUIVARIANCE & POLICY VALUE



Key Finding 2 (Model Comparisons & Welfare Dominance):

- **48.2% Customer Churn Prevented:** Tweedie and Hurdle maintain exact targeting stability, avoiding the 48.2% turnover observed when shifting a=1 to a=0.01 under log1p.
- **Highest Policy Value:** Tweedie achieves $0.8566 IPW policy value per targeted user at k=0.20, beating transformed baselines (+$0.12/user) on zero-heavy trials.

192 REPO UNIT TESTS

100% test pass rate across 16 suites verifying models, DGPs, metrics & oracle benchmarks.

rho = 1.0000 EQUIVARIANCE

Tweedie & Hurdle count-aware models maintain 100% targeting stability across all scales.

48.2% TARGETING CHURN

Top 20% targeted customers replaced under log1p merely by scaling dollars to cents.

4. Remaining Work: Execution Roadmap to Final Delivery

Completing simulation grids, empirical merges, manuscript writeup, and defense preparation

PHASE 1 | SIMULATION RUNS

Simulation Grid & Aggregation

- **Stage 1 Grid Completion:** Complete remaining replications of the full 6 x 4 x 5 factorial grid (92+ replications already logged in parallel background execution).
- **Error Ribbon Aggregation:** Aggregate cross-replication standard deviations across all 6 scenarios to render final multi-seed confidence bands for publication figures.
- **Heavy-Tail Stress Testing:** Evaluate Pareto tail index alpha in {1.5, 2.5} to demonstrate vulnerability of transformed models to high-spend outliers.
- **Summary Metrics:** Compile multi-seed tables for mean regret, Kendall's tau, and Top-k Jaccard overlap.

PHASE 2 | EMPIRICAL VALIDATION

Empirical Pipeline Merge & Criteo

- **Hillstrom Pipeline Integration:** Merge Member Q's real Hillstrom cross-validation evaluation pipeline into the main branch to unify team workflows.
- **Criteo 200k Full Pipeline Run:** Run full cross-fitted DR-Learner and Hurdle pipelines across the high-dimensional Criteo semi-synthetic benchmark (12 features, N=200,000).
- **Propensity Trimming Sensitivity:** Test whether extreme propensity weights (e(X) < 0.01 or > 0.99) amplify transformation sensitivity in Doubly Robust pseudo-outcomes.
- **Bootstrap Significance Tests:** Calculate 95% confidence intervals and p-values for difference in policy value between Tweedie vs log1p on Hillstrom test splits.

PHASE 3 | REPORT & SUBMISSION

Report Finalization & Saturday EOD

- **LaTeX Manuscript (report/main.tex):** Finalize Section 5 (Practical Decision Guide for Industry) and Appendix asymptotic proofs (a -> 0 and a -> inf). Verify 16 academic references and compile clean camera-ready PDF.
- **One-Command Reproduction Bundle:** Verify clean replication pipeline (make reproduce / run_all.sh) ensuring deterministic reproducibility across seeds.
- **GitHub Release & Code Freeze:** Push final commits and tag release on branch q/engine (https://github.com/pallaviB463/Rescaled.git).